Hanyu Meng

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Research Interests: Speech Enhancement, Target Speaker Extraction, Spatial Audio, and Music Information Retrieval

- **Robust Speech and Audio Front-ends:** Learnable and adaptive representations for speech enhancement and separation, with biologically inspired monaural and binaural front-end design.
- **Spatial Audio and Selective Hearing:** Binaural target speaker extraction, multi-speaker localisation and tracking, spatial speech enhancement, and room-acoustic parameter estimation.
- **Music Information Retrieval:** Automatic transcription of singing voices, neural instrument synthesis and timbre morphing, and generative models for music creation.

Education

University of New South Wales (UNSW) Sydney, Australia
Ph.D. in Electrical Engineering (Audio Signal Processing) *Jun 2022 – Sep 2026*

- Thesis: *Biologically Inspired Adaptive Front-Ends for Robust Spatial Machine Hearing*
- Research focus: Adaptive monaural and binaural front-ends for robust speech enhancement, target speaker extraction, multi-speaker localisation, and acoustic-context estimation.
- Supervisors: Prof. Eliathamby Ambikairajah and A/Prof. Vidhyasaharan Sethu

University of New South Wales (UNSW) Sydney, Australia
B.Eng. in Telecommunications (Honours) *Jul 2018 – May 2022*

- Honours: First Class Honours
- Thesis: *Singing Voice Separation by Embedding Accompaniment Repetitive Features into the Deep Neural Network*
- Supervisor: A/Prof. Vidhyasaharan Sethu
- Weighted Average Mark (WAM): 85.77/100 (top 1%)

Honours & Awards

The Australasian Speech Science and Technology Association (ASSTA) New Researcher Award	2026
The Australasian Speech Science and Technology Association (ASSTA) Conference Travel Award	2026
ISMIR Travel Grant Award	2026
ISMIR Financial Support Grant and Author Registration Waiver	2026
Asian Deans' Forum 2025, The Rising Stars Women in Engineering	2025
IEEE Signal Processing Society (SPS) Travel Grant	2025
UNSW 3 Minute Thesis (3MT) Runner-up Prize, School of Electrical Engineering and Telecommunications	2024
UNSW University International Postgraduate Award (UIPA)	2022
UNSW Taste of Research Scholarship	2021
UNSW Dean's Honour List	2021
UNSW Dean's Honour List	2019

Industry Working Experience

Meta – Reality Labs, Audio Team Cambridge, UK
Research Scientist Intern, Mentors: Dr Jesus M Alvarez and Dr Juan Azcarreta Ortiz *Jun 2026 – Dec 2026*

- Researching context-aware open-ear rendering (Near-end Listening Enhancement) for conversational focus in Meta smart glasses, supporting clearer target speech in complex acoustic scenes.

Sony Corporation, Audio Team Osaki, Tokyo, Japan
R&D Engineer Intern, Mentors: Tomohiro Sawada and Keiichi Taguchi *Jun 2025 – Jul 2025*

- Designed a user-configurable music style-transfer interface model that transforms pop tracks into Lo-fi versions for relaxation and study.
- Fine-tuned a pretrained diffusion model (Stable Audio Tool) for the music style-transfer system.

Sony Computer Science Laboratories Remote
Research Intern, Mentor: Taketo Akama *Jan 2025 – Jun 2025*

- Developed a diffusion-based model for instrument sound synthesis and timbre morphing.

- Implemented the model within the Music2Latent framework.

Dolby Laboratories, Advanced Technology Group Sydney, Australia
Research Intern, Mentors: Dr Jeroen Breebaart and Dr Jeremy Stoddard *May 2024 – Sep 2024*

- Investigated blind acoustic-context estimation from first-order Ambisonic recordings.
- Designed spectro-spatial features and neural back ends for estimating room-acoustic parameters.

Institute of Software, Chinese Academy of Sciences (ISCAS) Beijing, China
Research Assistant, Mentors: Prof. Changyou Zhang *Feb 2021 – Jul 2021*

- Developed a computer-vision prototype for detecting and localising vehicle defects in inspection imagery.

Academic Research Experience

Great Bay University Guangdong, China
Research Collaborator, Mentor: Dr Yaolong Ju *Feb 2026 – Present*

- Developing an end-to-end, part-aware transcription model for choral recordings with singing-voice assignment.
- One paper accepted at ISMIR 2026 Late Break Demo
- One paper submitted to ICASSP 2027
- Preparing a manuscript for submission to *IEEE/ACM Transactions on Audio, Speech, and Language Processing.*

Publications

Conference Papers

[1] **Hanyu Meng**, Vidhyasaharan Sethu, Eliathamby Ambikairajah, Qiquan Zhang, and Haizhou Li. BiEAR: A Human Auditory-Inspired Adaptive Binaural Front-end for Multi-Speaker Localisation and Distance Estimation. *Proc. Interspeech.* 2026. [Paper]

[2] **Hanyu Meng**, Vidhyasaharan Sethu, Eliathamby Ambikairajah, Qiquan Zhang, and Haizhou Li. Adaptive Per-Channel Energy Normalization Front-End for Robust Audio Signal Processing. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).* 2026. [Paper]

[3] **Hanyu Meng**, Jeroen Breebaart, Jeremy Stoddard, Vidhyasaharan Sethu, and Eliathamby Ambikairajah. Blind Estimation of Sub-band Acoustic Parameters from Ambisonic Recordings using Spectro-Spatial Covariance Features. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).* 2025. [Paper]

[4] **Hanyu Meng**, Qiquan Zhang, Xiangyu Zhang, Vidhyasaharan Sethu, and Eliathamby Ambikairajah. Binaural Selective Attention Model for Target Speaker Extraction. *Proc. Interspeech.* 2024. [Paper]

[5] **Hanyu Meng**, Vidhyasaharan Sethu, and Eliathamby Ambikairajah. What is Learnt by the LEArnable Front-end (LEAF)? Adapting Per-Channel Energy Normalisation (PCEN) to Noisy Conditions. *Proc. Interspeech.* 2023. [Paper]

[6] Hongjie Chen, **Hanyu Meng**, Huimin Zeng, Ryan A. Rossi, Lie Lu, and Josh Kimball. Variable-Length Audio Fingerprinting. *ACM Multimedia (MM).* 2026. [arXiv]

[7] Hongjie Chen, **Hanyu Meng**, Gautam Bhattacharya, Lie Lu, Josh Kimball, and Ryan A. Rossi. Auto-MatchCut: An Audio-Visual Retrieval Framework for Seamless Match Cutting. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).* 2026. [ICASSP]

[8] Zhanhong He, **Hanyu Meng**, Defeng Huang, and Roberto Togneri. Beyond Piano: Cross-Instrument MIDI Velocity Estimation via Differentiable SoundFont Proxies. *International Society for Music Information Retrieval Conference (ISMIR).* 2026. [arXiv]

[9] Zhanhong He, **Hanyu Meng**, Defeng Huang, and Roberto Togneri. Joint Estimation of Piano Dynamics and Metrical Structure with a Multi-task Multi-Scale Network. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).* 2026. [arXiv]

[10] Xiangyu Zhang, Daijiao Liu, Hexin Liu, Qiquan Zhang, **Hanyu Meng**, Leibny Paola Garcia, Eng Siong Chng, and Lina Yao. Speaking in Wavelet Domain: A Simple and Efficient Approach to Speed up Speech Diffusion Model. *Empirical Methods in Natural Language Processing (EMNLP).* 2024. [arXiv]

Journal Papers

[1] **Hanyu Meng**, Jeroen Breebaart, Jeremy Stoddard, Vidhyasaharan Sethu, and Eliathamby Ambikairajah. A Unified Deep Learning Framework for Estimating Acoustic Context Parameters from First-Order Ambisonic Speech Recordings. *EURASIP Journal on Audio, Speech, and Music Processing.* 2026. [DOI]

Late-Breaking/Demo Papers

[1] **Hanyu Meng**, Zhanhong He, Kaifeng Li, and Yaolong Ju. PawChorale: A Scalable Synthetic Dataset for Part-Aware Choral Transcription and Separation. *Late-Breaking/Demo Session at the 27th International Society for Music Information Retrieval Conference (ISMIR).* 2026. Accepted. [Project Page]

Under Review Manuscripts and Other Projects

[1] **Hanyu Meng**, Zhanhong He, Zixun Guo and Yaolong Ju. Toward Part-Aware Choral Transcription with Singing Voice Assignment. *Submitted to ICASSP 2027* [arXiv].

[2] **Hanyu Meng**, Hao Shi, Zhanhong He, Qiquan Zhang, Yang Xiao, Sungwoo Cho, Haowei Lou and Haizhou Li. Speaker-Consistent Grounding for Confusion-Resilient LLM-Based Target Speech Extraction. *Submitted to ICASSP 2027* [Project Page] [arXiv].

[3] Chunxi Wang, Maoshen Jia, Jing Zhou, **Hanyu Meng**, Bohua Ye and Wenyu Jin. Seeing Sound, Hearing Space: EchoBind for Multimodal Room Impulse Response Estimation. *Submitted to AAAI 2027*.

[4] Hongjie Chen, **Hanyu Meng**, You Zhang, Ryan A. Rossi, and Franck Dernoncourt. Last.Hot: Enhancing Music Session Recommendation Datasets with Musicological Genes, Audios, and Lyrics. *Submitted to ICASSP 2027*.

[5] Yang Xiao*, Tianyi Peng*, **Hanyu Meng** and Ting Dang. When Intent Arrives Late: A Benchmark for Full-Duplex Speech Models under Delayed Intent Revelation. *Submitted to ICASSP 2027* [arXiv].

[6] Zhanhong He, **Hanyu Meng**, Yaolong Ju. CASM: Context-Aware Semi-Markov Post-Processor for Beat Tracking. *Submitted to ICASSP 2027* [Project Page] [arXiv].

Teaching & Mentoring

University of New South Wales Sydney, Australia
Teaching Assistant *Feb 2021 – Dec 2024*

- Delivered laboratory demonstrations and assessed coursework across undergraduate signal-processing courses from first to fourth year (ELEC2134, TELE3113, ELEC4622, ELEC3104, and ELEC9123).
- Served as Head Demonstrator for TELE3113 and mentored students on digital signal processing projects.

University of New South Wales Sydney, Australia
Marker *May 2022 – Present*

- Assessed examinations, laboratory reports, quizzes, and project deliverables across TELE3113, ELEC3104, ELEC1111, and ELEC2134.

Academic Service

Reviewer

- **Journals:**
  - IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP) 2026
  - IEEE/ACM Transactions on Affective Computing (TAC) 2024
- **Peer-Reviewed Conferences and Workshops:**
  - IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2025–Present
  - ISCA Interspeech 2025–Present
  - ISMIR 2026
  - AAAI 2026
  - Spoken Language Technology Workshop (SLT) 2026

Volunteer and Organizing Service

- Served as an online volunteer for the ISMIR Conference 2024

Technical Skills

Speech & Audio: Speech enhancement, target speaker extraction, speech separation, audio representation learning, multi-speaker localisation and tracking.
Programming: Python (PyTorch, NumPy, pandas), TensorFlow, MATLAB, C++, C, VHDL, LaTeX, Markdown.
Languages: English (fluent), Chinese (native), Japanese (beginner).